

# SARBIX SCHOOL OPERATING SYSTEM

## Institutional Demonstration Playbook & Evaluator Pitch Guide

|                                                                        |                                                                                             |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Client / Institutional Target:</b> Leading K-12 Schools & Academies | <b>Target Audience:</b> Principals, Trustees & Academic Directors                           |
| <b>Version:</b> 1.0 Production Prototype (Next.js 15 + Supabase Cloud) | <b>Live Demo URL:</b> <a href="http://localhost:3000/login">http://localhost:3000/login</a> |
| <b>Test Student Dossier:</b> Aarav Patel (Grade 10-A, ADM-2026-0891)   | <b>Cloud Database:</b> Supabase PostgreSQL (ap-southeast-2)                                 |

### 1. Executive Value Proposition — Why Sarbix OS?

Traditional school software is fragmented across isolated vendors: one vendor for admissions, another for student attendance, an external portal for fees, and third-party tracking for transport. **Sarbix School OS solves this by unifying all 7 institutional stakeholders into a single, high-velocity, real-time operating system.**

|                                                                                                                                                      |                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unified Single Source of Truth</b><br>Every student, teacher, fee payment, and bus route is instantly synced in real-time across all roles.       | <b>Rapid Morning Attendance</b><br>Teachers complete daily roll calls in under 30 seconds; parents and students receive instant live verification.     |
| <b>Frictionless Fee Collections</b><br>Parents pay fees in 10 seconds via auto-generated UPI QR codes; accountants issue GST tax invoices instantly. | <b>Enterprise Audit &amp; Compliance</b><br>Complete cryptographic audit logging of every record modification for CBSE, ICSE, and state accreditation. |

### 2. Evaluator Role Matrix & 1-Click Fast Auth

On the login page (<http://localhost:3000/login>), evaluators can use the **Evaluator Role Switcher** to autofill credentials with one click. All demo accounts use the standard password: **Password@123**.

| Role        | Evaluator Email             | Password     | Key Responsibilities & Scope                                                                   |
|-------------|-----------------------------|--------------|------------------------------------------------------------------------------------------------|
| Principal   | principal@delhi.sarbix.edu  | Password@123 | Executive Command Center, holistic campus analytics, student dossiers, live Supabase editing.  |
| Teacher     | teacher@delhi.sarbix.edu    | Password@123 | Class 10A homeroom roster, rapid batch attendance submission, academic timetable.              |
| Student     | student@delhi.sarbix.edu    | Password@123 | Digital Student ID, synced live attendance tracker, class schedule, homework portal.           |
| Parent      | parent@delhi.sarbix.edu     | Password@123 | Aarav's parent (Vikram Patel): instant UPI QR fee pay, official PDF tax receipts, bus route.   |
| Accountant  | accountant@delhi.sarbix.edu | Password@123 | Institutional fee ledger, create student invoices, reconcile bank collections, print receipts. |
| Transport   | transport@delhi.sarbix.edu  | Password@123 | Fleet telemetry, route waypoints, driver manifests for Route 04 (South Delhi Express).         |
| Super Admin | superadmin@sarbix.edu       | Password@123 | Multi-campus governance, role permission matrix, cryptographic compliance audit logs.          |

### 3. Pitch Walkthrough: End-to-End Management of Student Aarav Patel

To give school leadership an unforgettable demonstration, run this cohesive story centered around a real enrolled student: **Aarav Patel (ID: std-001, Class: Grade 10-A, Admission: ADM-2026-0891).**

#### Step 1: Principal Command Center & Supabase Cloud Management

**Log in as:** principal@delhi.sarbix.edu | **Destination:** /students/std-001

- **Inspect Dossier:** View Aarav's complete academic profile, attendance standing (94.8%), blood group (O+), and assigned bus route (Route 04).
- **Live Supabase Edit:** Click the purple '**Manage Student (Supabase)**' button. Notice the live indicator showing connection to Supabase PostgreSQL (Project: debhfqjgqmlmujrfyrny).
- **Modify Record:** Update his fee balance, emergency contact, or section, then click '**Save to Supabase**'. The update synchronizes to the cloud database with live confirmation.

#### Step 2: Morning Roll Call & Rapid Attendance Sync

**Log in as:** teacher@delhi.sarbix.edu | **Destination:** /attendance

- **Open Homeroom:** Select Class Grade 10, Section A. Aarav Patel appears at Roll #10A-14.
- **Mark Attendance:** Toggle Aarav to 'Present' (or 'Late') with one click.
- **Submit Roster:** Click '**Submit Roster & Sync**'. Attendance records are instantly persisted to the cloud, updating campus welfare statistics.

#### Step 3: Student Learning Hub & Digital ID

**Log in as:** student@delhi.sarbix.edu | **Destination:** /student

- **Digital ID Badge:** Aarav views his photo, QR-coded admission ID, and active status.
- **Live Attendance Tracker:** Observe the real-time attendance indicator showing that his homeroom teacher verified him this morning.
- **Academic Timetable:** Check his personalized daily class schedule, upcoming tests, and assignments.

#### Step 4: Parent Portal, UPI QR Fee Payment & Tax Receipts

**Log in as:** parent@delhi.sarbix.edu | **Destination:** /parent

- **Aarav's Welfare Overview:** Vikram Patel (father) monitors Aarav's attendance and academic progress.
- **Instant Dynamic UPI QR:** Click '**Pay Fees via QR**' on the ₹18,500 pending invoice. An instant UPI QR code modal opens with pre-populated VPA and invoice number.
- **Official PDF Receipt:** Click '**Download Tax Receipt**' to view, inspect, and print a fully formatted institutional fee receipt with GST breakdown.

#### Step 5: Institutional Finance Ledger & Invoice Issuance

**Log in as:** accountant@delhi.sarbix.edu | **Destination:** /finance

- **Fee Ledger:** View total collections, pending balances, and search Aarav Patel's ledger.
- **Generate New Invoice:** Click '**Create New Invoice**', select Aarav Patel, enter fee head and amount, and submit. The invoice is written directly to Supabase and instantly visible to his parents.

#### Step 6: Fleet GPS Telemetry & Student Bus Manifest

**Log in as:** transport@delhi.sarbix.edu | **Destination:** /transport

- **Active Fleet Status:** Monitor 8 active school buses across New Delhi routes.

- **Aarav's Commute:** Inspect **Route 04 (South Delhi Express)**. Verify driver contact, GPS coordinates, and student boarding roster.

## Step 7: Multi-Campus Governance & Cryptographic Audit Trail

**Log in as:** superadmin@sarbix.edu | **Destination:** /settings

- **RBAC Matrix:** Inspect granular permissions across Delhi Central, South, and Rohini campuses.
- **Audit Trail:** Verify the audit logs recording every action performed during the demo (student edits, fee generation, attendance updates) with timestamps and actor IDs.

## 4. Enterprise Database & Cloud Infrastructure (Supabase)

Sarbix School OS is built on an enterprise hybrid architecture that combines a high-speed local institutional cache with a live **Supabase Cloud PostgreSQL** database.

| Component            | Specification & Live Status                                                  |
|----------------------|------------------------------------------------------------------------------|
| Supabase Project Ref | debhfgjqgmImujrfyrny (ap-southeast-2)                                        |
| Database Engine      | PostgreSQL 17.6 Enterprise Managed Instance                                  |
| API Layer            | REST API v1 + Row Level Security (RLS) + JWT Auth                            |
| Core Cloud Tables    | students, attendance_records, fee_invoices                                   |
| Offline Resilience   | Automatic local fallback ensures zero disruption if school internet flickers |
| Live Verification    | Verified live: student read/write, fee invoice creation, and attendance sync |

## 5. School Pitch Talking Points & Objection Handling

**Q: Will teachers find it difficult to adopt?**

A: No. The interface is optimized for rapid mobile touch. Morning roll call requires only toggling present/absent and hitting submit, taking less than 30 seconds per classroom.

**Q: Can parents pay school fees easily without downloading complicated apps?**

A: Yes. Parents can open the portal on any smartphone, view their itemized invoice, and click 'Pay Fees via QR' to scan with Google Pay, PhonePe, or Paytm instantly.

**Q: What happens if school WiFi goes down during morning roll call?**

A: Sarbix OS features an offline-first resilient architecture. Attendance is queued locally and syncs back to Supabase cloud automatically as soon as connection is restored.

**Q: Can we customize permissions for our vice-principals and department heads?**

A: Yes. The system has a granular Role-Based Access Control (RBAC) matrix that allows institutional admins to configure custom view/edit rights per campus or department.

**Q: How does this help with school accreditation (CBSE / ICSE / IB / Cambridge)?**

A: All attendance rosters, grade sheets, student disciplinary notes, and fee transactions are timestamped with cryptographic audit trails ready for one-click compliance export.

**Institutional Evaluator Next Steps:**

1. Launch `http://localhost:3000/login`
2. Click through the 7 roles using the 1-Click Evaluator Presets.
3. Walk through student **Aarav Patel (std-001)** from Principal edit → Teacher attendance → Parent QR payment.
4. For institutional deployment inquiries, contact **Sarbix Growth Agency**.